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CSAS117 Cryptography and Network Security
Assignment - 03 - Set 03

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1) a) Purpose of HMAC:

1. Uses a secret key with a hash function.
2. Provides message authentication.
3. Ensures message integrity.
4. Prevents unauthorized modification of messages.
5. More secure for authentication than using a plain hash.

b) HMAC-SHA-256 Calculation:

Given:

- Message: HELLO
- Key: SECRET
- Hash: SHA-256

Steps:

1. Convert the key SECRET into bytes and pad it to 64 bytes.
2. XOR the padded key with ipad (0×36).
3. Append the message HELLO and apply SHA-256 to get the inner hash.
4. XOR the padded key with opad ($0 \times 5C$).
5. Append the inner hash and apply SHA-256 again.
6. The final 256-bit value is the HMAC.

Formula:

$$\text{HMAC} = \text{SHA256}[(K' \oplus \text{opad}) \parallel \text{SHA}[(K' \oplus \text{ipad}) \parallel \text{message}]]$$

Thus, the client sends the message along with HMAC and the server uses the shared key to verify it.

c) Why not MD5:

Reasons:

1. MD5 is cryptographically broken.
2. It has known collision attacks.
3. It produces only a 128 bit hash.
4. It is unsuitable for modern security critical systems.
5. H-MAC-SHA 256 is preferred.
6. Attackers can exploit its weakness.

d)

MAC vs Digital Signature:

MAC - Authentication + Integrity

Digital Signature $\rightarrow$ Authentication + Integrity + Non repudiation.

MAC

Uses a shared key
Provides authentication and integrity

Does not provide non repudiation

Mainly for

MAC

Uses a shared secret key
Provides authentication and
integrity

Does not provide non-
repudiation

Mainly used between
trusted parties

Digital Signature

Uses private and public keys

Provides authentication and
Integrity

Provides non-repudiation.

Can be verified by anyone with
the public key.

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